



# MOUNTAIN VIEW STREAMING PLATFORM

## MOVIE RECOMMENDER SYSTEM ANALYSIS

LEVERAGING MATRIX FACTORIZATION AND CONTENT ANALYSIS TO REDUCE CHURN THROUGH PERSONALIZED CONTENT SURFACING.

# BUSINESS OVERVIEW

## The problem

### User Churn

Mountain View Platform faces a critical challenge: users are leaving the platform when recommended content fails to match their unique viewing preferences.

## The goal

### Personalization

Reduce churn by personalizing the home screen specifically for returning users who have provided at least 10 historical ratings.

## The strategy

### Top-5 Delivery

Surface the Top-5 personalized movies. For new users, an onboarding survey of 5-10 popular titles mitigates the cold-start problem.

# BUSINESS OBJECTIVES AND SUCCESS CRITERIA

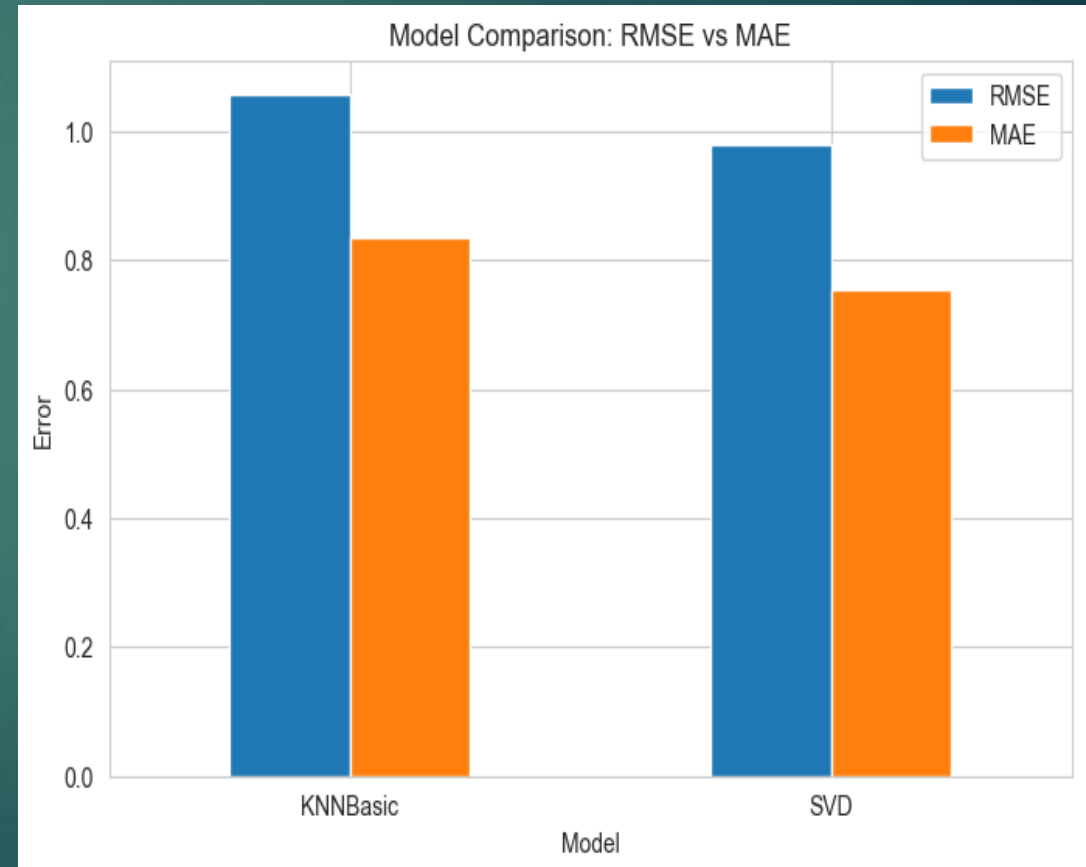
SVD outperforms KNN Basics

## Business objectives

- .Increase user retention
- Increase User engagement
- Balance popular content with personalized recommendations
- 4.Support new and returning Users

## Success Metrics

- SVD outperforms KNNBasic
- Hyperparameter tuning improves SVD
- Generate Top-5 personalized recommendations
- Address cold-start users

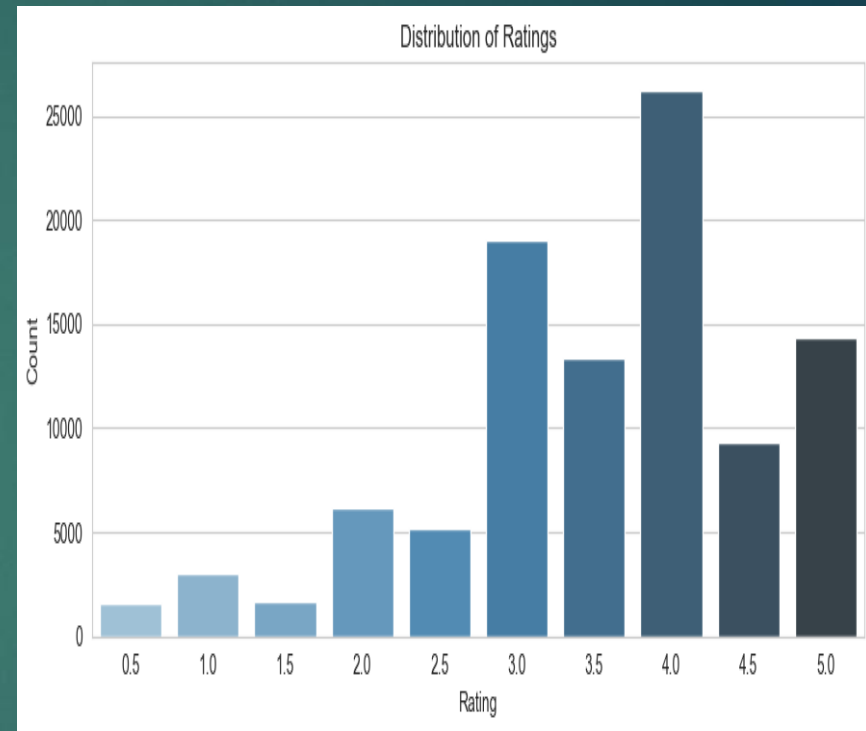


# DATA INSIGHTS

32M

TOTAL RATINGS.

- Drama and Romance, Comedy and Romance are the most frequent genre pairings.



Ratings range from 0.5 to 5.0, with a mean of 3.5. Most users tend to rate positively.

# MODEL PERFORMANCE

0.67

## RECALL@ 5

The system successfully identifies a high proportion of movies that users would actually like

0.15%

## PRECISION@5

Lower precision reflects the challenge of "correct" niche predictions being buried by popular content.

## Popularity Bias & List Quality

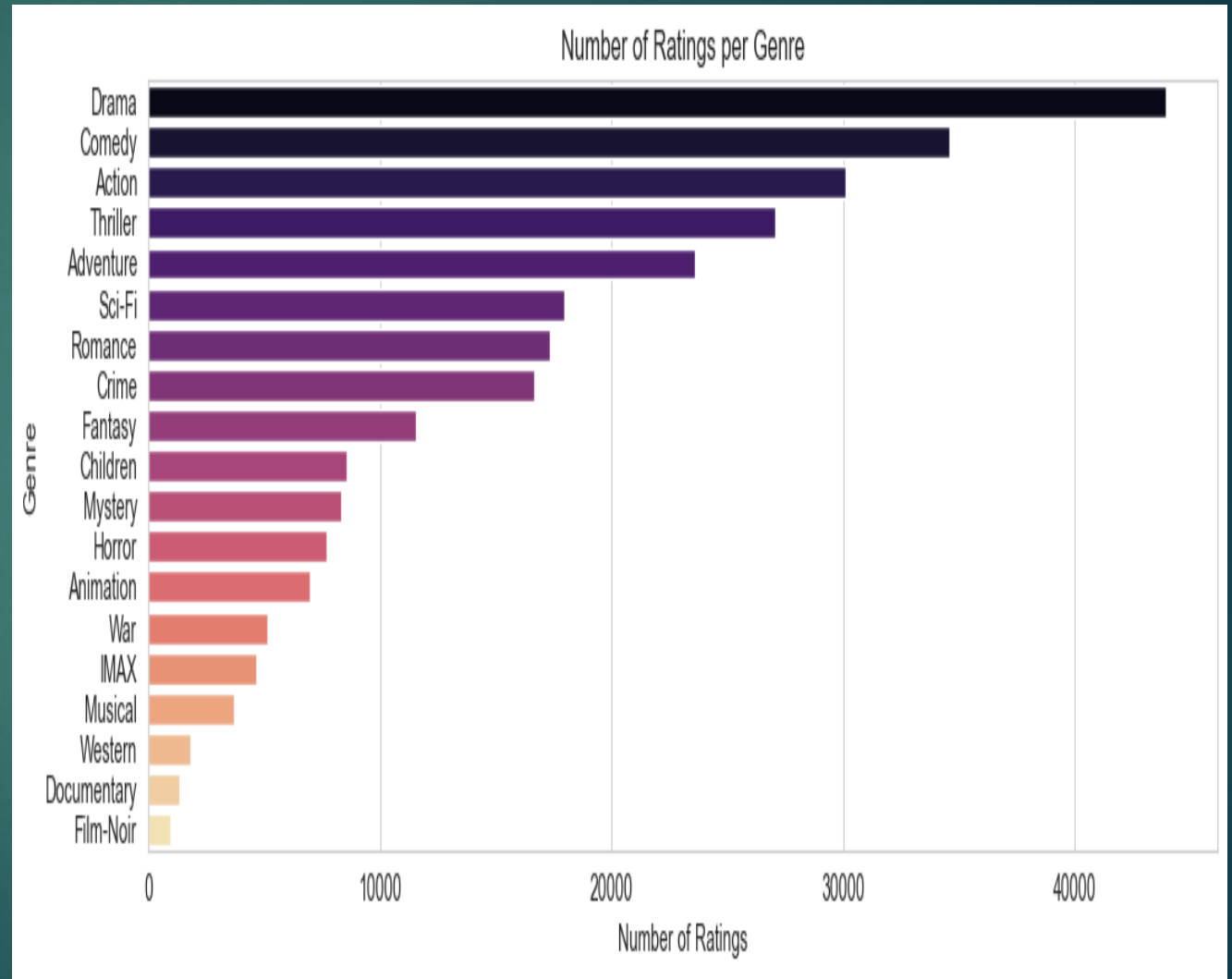
- **The Bias:** Collaborative filtering tends to favor widely-liked movies, often missing specific individual tastes in the Top-5 list.
- **Recommendation Gap:** While recall is high, the low precision indicates that the "hits" are often not in the top-most positions.
- **Strategic Need:** Future iterations require re-ranking strategies to penalize over-exposed popular content and boost diversity

# GENRE DOMINANCE AND CONTENT DIVERSITY

**Genre Concentration:** Drama and Comedy heavily weight the dataset.

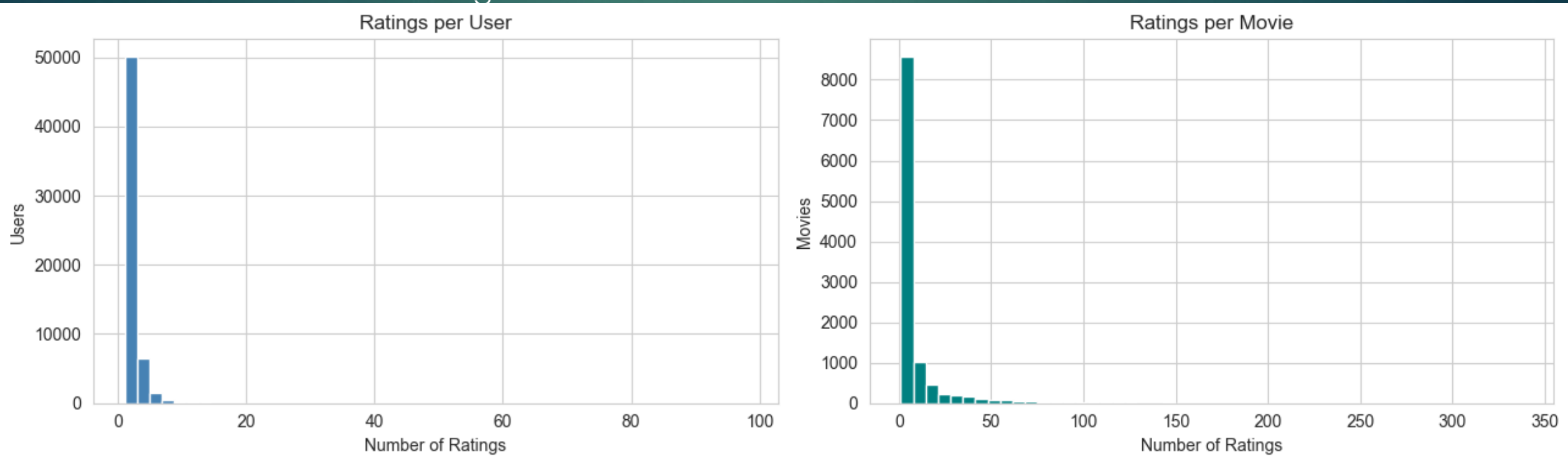
**Diversity Gap:** Standard collaborative filtering may overlook niche genres like Documentary or Film-Noir without active intervention.

**Strategic Balancing:** Using genre-based cosine similarity helps maintain diversity, especially during the critical cold-start phase.



# The Long-Tail Problem

- **Right-Skewed Data:** A small number of "blockbuster" movies receive the vast majority of ratings, dominating the model.
- **Power User Bias:** A minority of users contribute most of the data, potentially skewing recommendations toward their specific tastes.
- **The Long Tail:** Niche movies lack sufficient interaction data, making them difficult for collaborative filtering to surface.





# Solving the Cold-Start Dilemma

## Returning Users

SVD Model: Active when user has  $\geq 10$  ratings

Personalization: Leverages latent features from historical interactions.

Output: Highly specific Top-5 movie predictions

## New Users (1-9 Ratings)

Content-Based: TF-IDF vectorization on movie genres

Similarity: Recommends items similar to the few movies already liked

Bridge: Maintains quality until enough data is gathered for SVD.

## Zero-Data Users

Popularity Fallback: Surfaces highest-rated movies with  $> 50$  ratings

Conversion: Goal is to prompt the first 10 ratings quickly.



# Conclusion: Path to a Scalable Recommender

- ✓ SVD Superiority: Matrix factorization consistently outperforms KNN in handling sparse movie rating data.
- ✓ Full Dataset Training: Moving from 100k samples to the full 32M ratings is expected to close the RMSE/MAE gap
- ✓ 100% Coverage: The combination of SVD and Content-Based models ensures every user receives recommendations.
- ✓ Richer Metadata: Incorporating cast, directors, and plot keywords to improve cold-start accuracy.
- ✓ Diversity Re-ranking: Implementing penalties for over-exposed popular content to boost niche discovery.

THANK YOU

Q&A